

Stress Detection during Social Interactions with Natural Language Processing and Machine Learning

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Abstract - The field of Natural Language Processing (NLP) is witnessing a growing recognition of the significance of investigating stress-related phenomena. This research study proposes a new framework to address the diagnostic challenges associated with mental health conditions such as anxiety, depression, and stress, which can worsen if left unaddressed. Within this research study, social media postings and comments are used to enhance sentiment and emotion analysis for assessing individuals' stress levels. Specifically, text analysis of social media data, including Twitter tweets, serves as a pivotal application. The proposed system is implemented by employing a combination of Machine Learning (ML), Natural Language Processing (NLP), and Deep Learning (DL) methodologies. The proposed system utilizes Convolutional Neural Network (CNN), BERT transfer learning techniques, as well as Machine Learning (ML) algorithms such as Naive Bayes, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). This study aims to predict five distinct emotional categories: "Happy," "Angry," "Depression," "Sad," and "Fear." Remarkably, the proposed system has achieved an accuracy and F1 score exceeding 93%, outperforming the performance of other existing human emotion recognition systems.

Keywords: Machine learning, deep learning, BERT, CNN, stress, NLP, social media.

I. INTRODUCTION

Everyone has occasionally been stressed, regardless of gender, age, or culture. Our reaction to stress is the most important component, and it will be highly important [1]. Stress can be advantageous or harmful. In particular, the COVID-19 outbreak caused lockdowns and excessive protracted social isolation, which put almost everyone in a condition of stress. Because of these limitations, people in the epidemic era spend a lot of time on social media [2, 3]. These media are helpful in a number of disciplines, including product research, political campaigns, advertising, and voting procedures for large-scale festivals. The most widely used technique for gathering and evaluating data from some websites, blog postings, and virtual entertainment platforms to help associations make defensible decisions is social media analytics (SMA). Utilizing social media has become a common practice in today's environment. SMA is utilized for impact, assessment, and opinion mining these days. It can be used,

nevertheless, to evaluate a person's emotional state.

However, measuring and analyzing social elements can be challenging. Big Data, ML, and NLP techniques must be used to analyze large amounts of often unstructured data in order to draw conclusions regarding stress or other mental health issues. Research has demonstrated that healthcare systems assess and quantify social interaction to identify stress and depression using constructive SMA in combination with other health indicators [4].

Stress can be the source of both physical and emotional problems. It usually gets triggered by anything or anyone that makes you angry, irrational, or restless. Adjustments in everyday habits and desires can lead to stress, a mental health disorder that affects both the body and the mind [5].

Through the application of NLP, we are able to deduce emotional content from social media posts. NLP uses Computing techniques to examine etymological information found in printed materials such as texts or articles [6]. Computers find it challenging to read documents since of their disorganized layout. In general, this strategy facilitates text processing. Experiments confirm the claim that the BERT model performed better for this task than other popular approaches without requiring a complex design. After that, we divide feelings into five categories using the BERT model. In light of the aforementioned, an automated system that can promptly detect stress, depression, and anxiety and alert the person is required in order to treat the issue and improve the person's quality of life. NLP and machine learning applications are utilized to enhance mental wellness.

Research Objectives

- To gather and comprehend the behavior of social media users across various social media platforms.
- To use deep learning and machine learning methods to detect mental health problems in individuals.
- To automatically identify mental health issues based on social media posts, such as depression, anxiety, tension, and sorrow.

II. LITERATURE REVIEW

Computer-assisted emotion identification, analysis, and application have garnered a lot of attention lately, especially on social networks. [7] Stress and psychology: connections [8] Personality qualities could be a crucial aspect to take into account. As an illustration, shows that Daily44 is trustworthy information source [9]. Metrics based on behavior can be used to accurately identify stress. Users using mobile phones render services. Traditional categorization techniques and text-based linguistic elements are used in many tweet-level social media emotion analysis research [10]. Some studies looked into the problem of emotional transmission on social networks and found that users were more likely to share anger than joy, which meant that negative emotions might travel farther and faster.

While emotional states or mental strain are usually more stable and change over time, tweet-level emotion detection captures the instantaneous emotion expressed in a single tweet. In the past several years, a lot of investigation has been done on user-level recognition of emotions in social networks [11]. In order to evaluate people mental health states from social media, our latest research [12] suggests using a deep convolution network to learn user-level expression on successive tweet series over a predetermined period of time. Few studies enhanced sentiment analysis at the user level on Twitter by the integration of social interactions, drawing inspiration from the homophile concept [13]. While considerable research has been done on user-level emotion recognition, it has not yet been investigated how social ties affect psychological stress states or how to use this information in stress detection.

One of the key components of social media networks is social networking. Researchers are now focusing on exploring the ways to leverage social interaction data to improve the efficacy of social media evaluation. Few researchers looked into how users' views and actions were influenced by their social connections and discovered that interactions on Twitter can result in well-thought-out conclusions [14]. Some studies predicted users' feelings based on comments left on photos uploaded to Flickr [15]. On the other hand, this research ignored the underlying structural elements, such as the relationships between users, and instead concentrated only on the textual comment material that is part of social interactions.

III. METHODOLOGY

3.1 Dataset

To prepare the output labels for every text or sentence that is collected, the Text2Emotion library is utilized. The generated dataset consists of around 25K distinct Twitter posts divided into five categories: "Happy," "Angry," "Surprise," "Sad," and "Fear." There are variations in the quantity of cases across all datasets. There is 1228 ID in the Dataset-I. There are 21429 sentiment levels in dataset II, and 2343 sentiment text in dataset III.

3.2 Preprocessing

To train the deployed models, a text corpus is collected from three different social media networks. The Text2Emotion software is used to generate the dataset's output labels. The prepared dataset has two properties, Text and Catg, and a total of twenty -five thousand unique occurrences. Following the completion of the dataset collection, the data from social media posts is subjected to a number of pre-processing procedures, using stop-word elimination, special symbol removal, and sentence tokenization. The performance of the applicable algorithms is essentially improved by all of this text preprocessing.

3.3 Preparation of Output labels

The text2emotion package serves as the foundation for the preparation of the dataset's output labels. Finding appropriate emotions concealed in text data is the main objective of the Python library Text2emotion. Essentially, it yields five human emotions along with confidence scores; for our label class, the feeling we are looking at here has the most confidence. Following the application of the output labels to the several social media posts, the datasets are combined into one.

3.4 Converting Text to Vector

Text datasets cannot be used to directly feed text data into ML and DL algorithms for training. There must be some text to vectorize form conversion. In order to convert the text into a vectorize format for use as input in the training of the models, three methods are applied to the data: TF-IDF, Word2Vec and Bag of Words (BoW). Word2vec represents a neural network structure that can be instructed to produce word embeddings on a supervised classification task. Tensor Flow (TF) is a DL framework created by Google Brain. To train the DL classifier, text classification is done using Tensor Flow.

3.5 Algorithms Used

Following the dataset's vectorization, the corpus was utilized to train CNN deep learning models, transfer learning via BERT, and CNN paired with BERT for machine learning (like Naïve Bayes, KNN, and SVM). The previously stated models are trained with and without Word Sense Disambiguation (WSD) approaches in order to predict the mental health of social media users.

- *Naive Bayes*: A statistical classifier formula is the Naive Bayes's classifier. It is based on the Bayesian Theorem. Given its simplistic design and naive idea, it can be used to incredibly complex problems. The foundation of this classifier is the idea that no property depends on any other attribute.
- *Support Vector Machine*: In order to generate regression and classification models, one useful Supervised ML technique is the Support Vector Machine (SVM) algorithm. It is straightforward and efficient. SVM's primary objective is to build the hyper plane and split the dataset into distinct groups so that data mining and categorization may be completed effectively. SVM can be used to solve practical issues like identifying tension in tweets.
- *KNN*: Similar objects are implied to be near together by the KNN algorithm. Stated differently, relevant objects are grouped together.
- *Convolutional Neural Network Model*: Regularized multilayer perceptrons make up CNNs. Fully connected networks are commonly used to describe multilayer perceptrons, where each neuron in a layer is tied to every other layer's neuron. Three different types of layers make up CNN: input, hidden, and output layers.
- *BERT*: BERT is a language processing (NLP) artificial intelligence platform that is available to the public. BERT provides context through surrounding information to assist computers in deciphering the meaning of emotive language in text.

3.6 Output

This is the last phase of the architecture. It concludes by listing the possible reactions to the social media post as follows: "Happy," "Angry," "Surprise," "Sad," and "Fear." The fraction of accurate predictions compared to all tweets is utilized to calculate accuracy.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

The percentage of correctly categorized positive tweets relative to the total number of positive tweets is used to assess precision.

$$Precision = \frac{TP}{TP + FP}$$

Recall is divided by the total number of tweets to get the proportion of positively classified tweets.

$$Recall = \frac{TP}{TP + FN}$$

The weighted mean of recall and precision is known as the F1-score,

$$F1 - score = \frac{2 \cdot precision \cdot recall}{precision + recall}$$

Figure 1 below displays the process's proposed framework in its entirety.

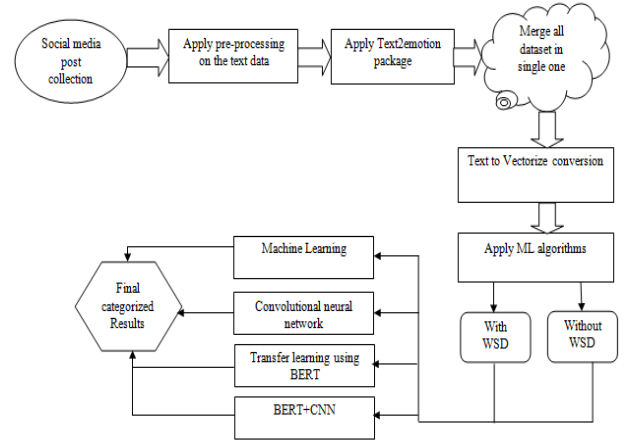


Figure 1: Proposed framework

IV. RESULTS & DISCUSSION

Several classifiers were used in the suggested system to identify stress and other mental states based on the input set that was provided. We are forecasting five emotion categories— Angry, Surprise, Happy, Fear, and Sad—as a result of the suggested approach. Mental health professionals are able to determine if a person's mental state is stressed or relaxed based on the automatic prediction of the categories. Table 1 displays the various classifiers' performance.

TABLE 1: RESULTS OF THE MENTAL HEALTH DETECTION ALGORITHM SUMMARIZED

Models	Accuracy	F1 Score	Precision	Recall
Naïve Bayes	84.48	86.37	84.26	86.83
KNN	76.98	76.85	74.63	76.58
SVM	89.36	87.24	85.12	88.74
BERT	92.86	91.52	89.31	91.79
CNN	81.37	79.64	77.42	80.64
CNN + BERT	96.84	95.37	93.15	94.59

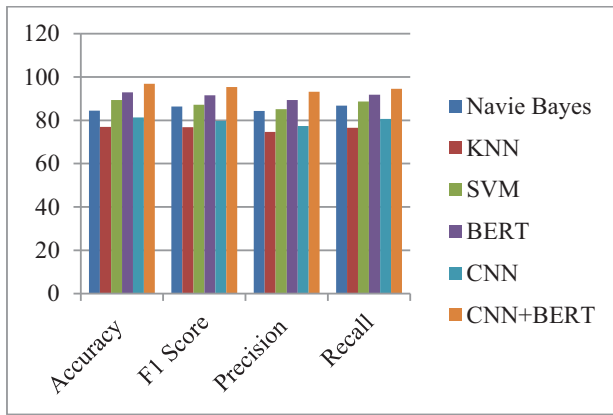


Figure 2: Comparative Analysis of Implemented models

The accuracy value comparison for each algorithm is displayed in Figure 2. The F1-scores of 95.37 and accuracy demonstrated by the CNN and BERT combo outperformed those of the other five approaches. Similar outcomes to the CNN + BERT combo classifier was also displayed by the KNN and BERT. The confusion matrix for BERT + CNN is shown in Figure 3 for each of the five emotional classes.

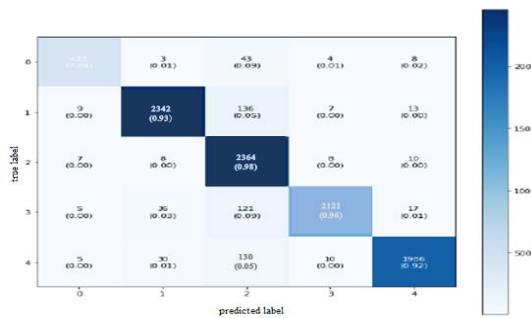


Figure 3: CNN+BERT Confusion Matrix

Figure 4 displays the training and validation graph for the LSTM model accuracy; it is clear from the graph that the model is not over fitted.

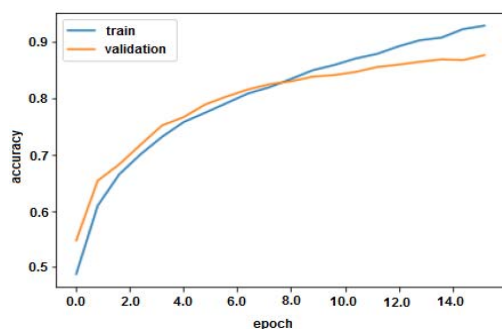


Figure 4: Accuracy-related training validation

The fact that the validation dataset is close to the model's training data indicates that the model is fitted correctly. Figure 5 displays the training and validation graph for the LSTM model loss.

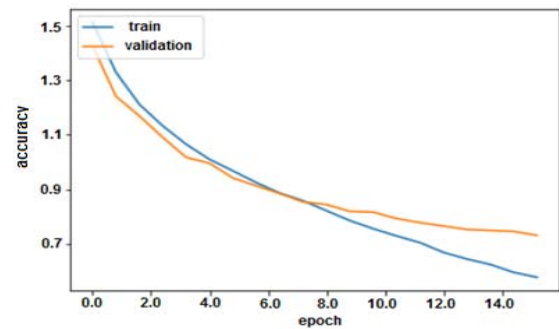


Figure 5: Validation of Training Concerning Loss

Additionally, this graph shows that the model is fitting the data correctly. As we can see, the model is not over fitted because the validation dataset is close to the training data. The suggested approach aims to lessen the issue of mental health problems associated with stress, anxiety, and depression while offering a clever way to detect them early on and prevent them from getting worse. By integrating NLP with ML and DL techniques, the proposed system would be able to automatically diagnose mental health issues (such as anxiety, stress, sadness and depression) depend on social media posts. Additionally, the algorithm picks up on the patterns of how online users publish and exchange ideas on social networking sites.

V. CONCLUSION

The proposed model has the ability to improve people's quality of life by identifying mental health disorders in individuals by utilizing ML and DL classification algorithms. Using a publically available dataset, this study has successfully deployed many ML and DL classification models to achieve faster and more accurate human emotion recognition. With the proposed model, the mental health experts will be able to identify mental health conditions early and with greater speed and accuracy by leveraging an automatic stress analysis framework. In the near future, more precise outcomes can be obtained by pre-processing the text thoroughly and combining different optimization strategies to minimize information loss.

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